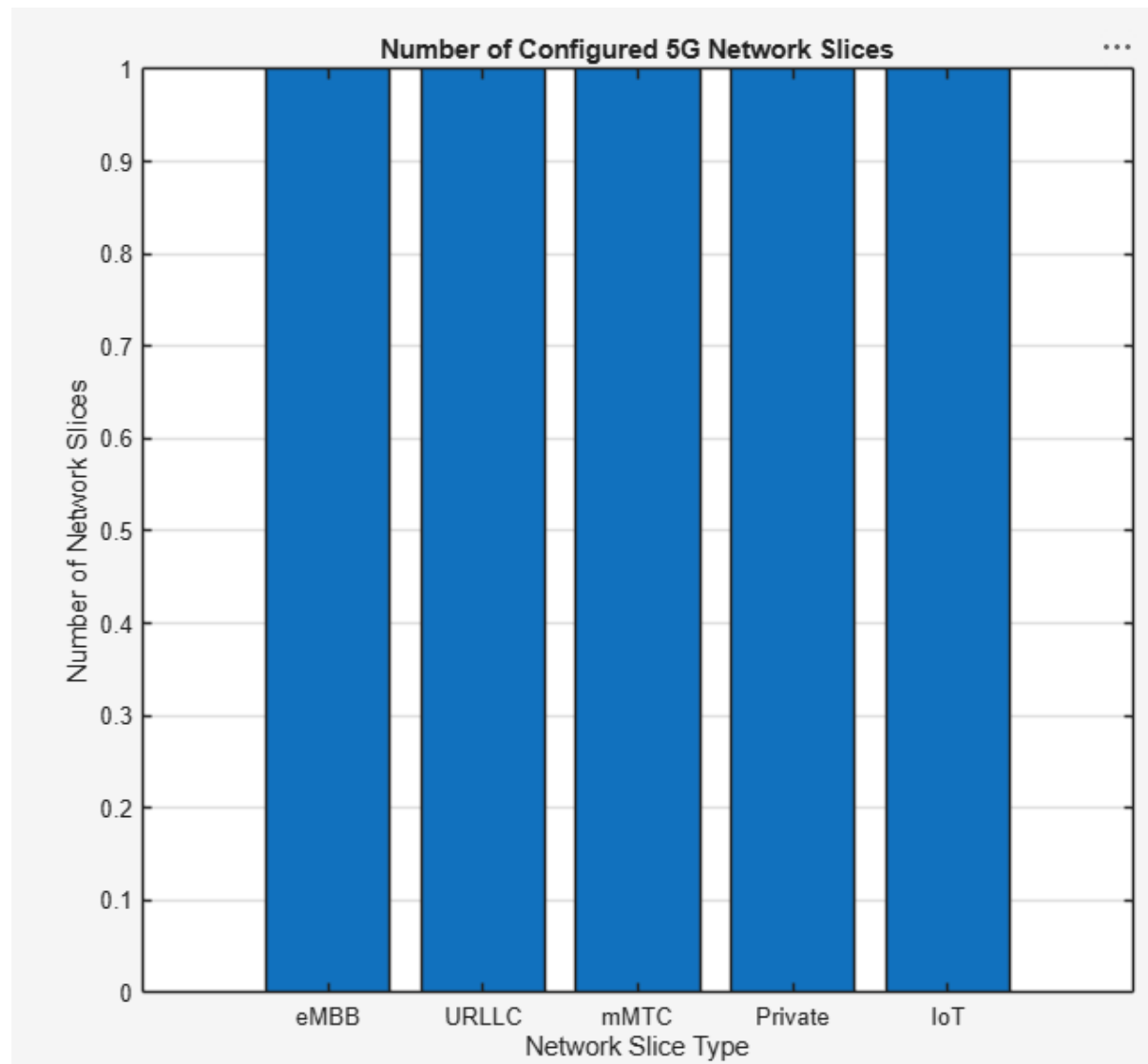


Ex.No.10 5G Network Slicing Environment Using MATLAB

1: Matlab Code and Output

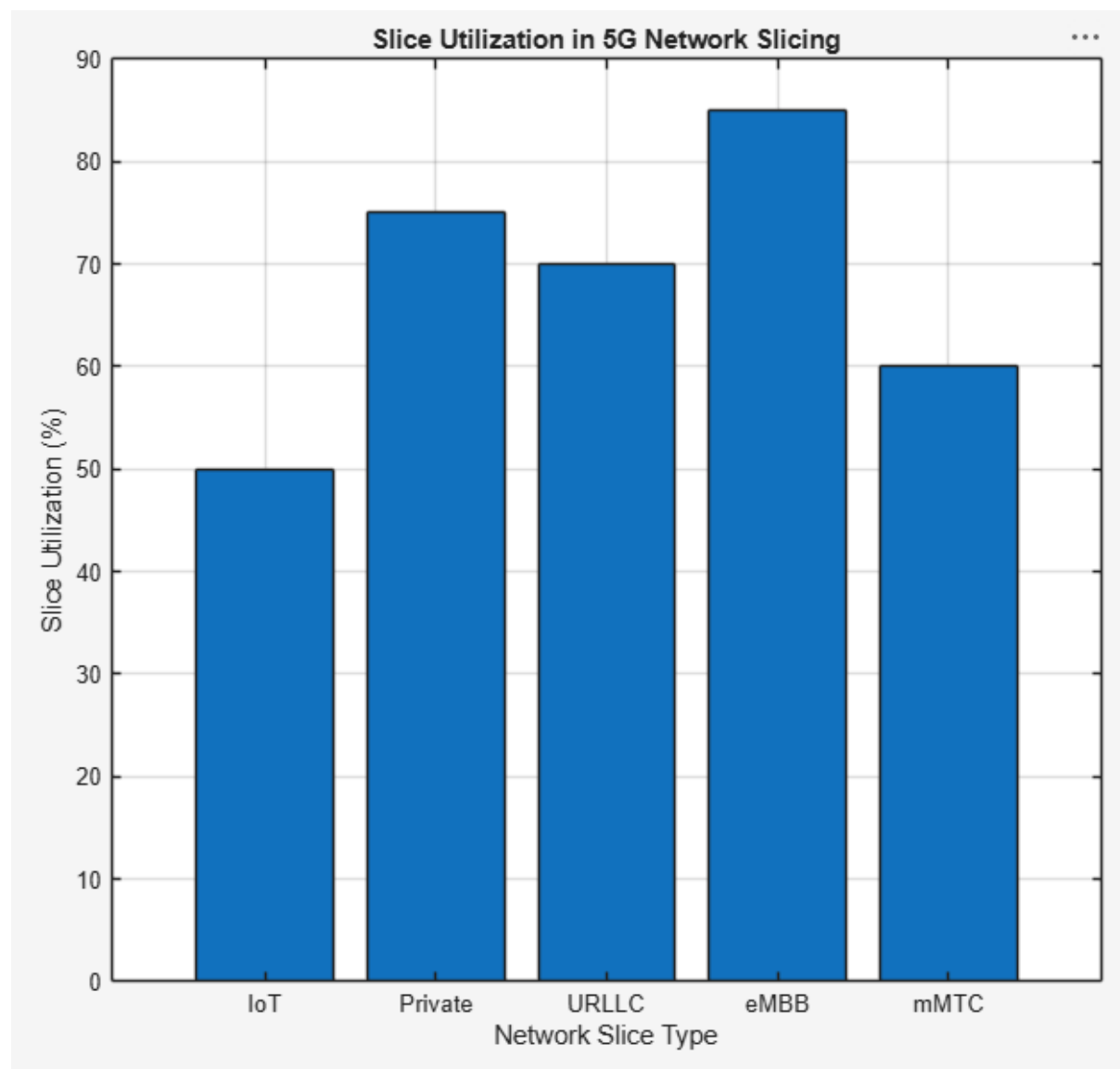
/MATLAB Drive/slotb10.m

```
1   clc;
2   clear;
3   close all;
4   slices = {'eMBB','URLLC','mMTC','Private','IoT'};
5   count = [1 1 1 1 1];
6   figure
7   bar(count)
8   % One instance of each network slice
9   grid on
10  set(gca,'XTickLabel',slices)
11  xlabel('Network Slice Type')
12  ylabel('Number of Network Slices')
13  title('Number of Configured 5G Network Slices')
14  "
```



Objective 2: Matlab Code and Output

```
15 slices = categorical({'eMBB','URLLC','mMTC','Private','IoT'});
16 utilization = [85 70 60 75 50];
17 figure;
18 bar(slices, utilization);
19 grid on;
20 xlabel('Network Slice Type');
21 ylabel('Slice Utilization (%)');
22 title('Slice Utilization in 5G Network Slicing');
23
```



Objective 3: Matlab Code and Output

```
24 slices = [1 2 3 4 5];
25 latency = [25 5 40 15 60];
26 figure;
27 plot(slices, latency, '-o','LineWidth',2,'MarkerSize',8);
28 grid on;
29 xticks(slices);
30 xticklabels({'eMBB','URLLC','mMTC','Private','IoT'});
31 xlabel('Network Slice Type');
32 ylabel('End-to-End Latency (ms)');
33 title('5G Network Slicing: End-to-End Latency');
34 ylim([0 70]);
```

